

Distribution and Mixing Properties of Modular Exponentiation

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Let N, g be positive integers. Define the **exponential map** $\gamma_{N,g} : \mathbb{N} \rightarrow \mathbb{N}$ by

$$\gamma_{N,g}(x) = g^x \pmod{N}.$$

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- Despite its widespread use, many properties remain unproven.
 - E.g. no efficient method is known for undoing the map.
- This motivates our study into the randomness of the **exponential map**.

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Definition

Call g a **primitive element** (or **primitive root** or **generator**) modulo p , if $\text{ord}_p(g) = p - 1$. Equivalently, the powers of g modulo p generate all non-zero residues modulo p ,

$$\{g^x \pmod{p} : 1 \leq x \leq p - 1\} = \{1, 2, \dots, p - 1\}.$$

Definition

We write $f \ll g$ if there exists constant $C, N > 0$ such that $|f(x)| \leq C|g(x)|$ for all $x \geq N$. Alternatively, this is denoted by

$$f(x) = O(g(x)).$$

We say $f = o(g)$ if

$$\lim_{x \rightarrow \infty} \left| \frac{f(x)}{g(x)} \right| = 0.$$

We say g is an **asymptotic formula** for f if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1.$$

Denote this by $f \sim g$.

Problem 1 - Almost Fixed Points

Definition

Let p be a prime, and g be a **generator** modulo p . For $h \in \{0, 1, \dots, (p-1)/2\}$, define $M_{p,g}(h)$ as the number of solutions to

$$g^x \equiv x + y \pmod{p},$$

for $x \in \{1, \dots, p-1\}$ and $y \in [-h, h]$.

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- “How often is g^x close to x modulo p ?”
- What is the behaviour of $M_{p,g}(h)$?
 - If $\gamma_{p,g}$ behaves like a random map, we expect $M_{p,g}(h) \approx 2h$.

Existing Results - Fixed Points and Short Cycles

Recall

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Definition

Let p be a prime, and g be a generator modulo p . For positive integer k , define $T_{p,g}(k)$ be the number of solutions in $x \in \{1, \dots, p-1\}$ to

$$\underbrace{\gamma_{p,g}(\dots \gamma_{p,g}(x) \dots)}_{k \text{ times}} = x.$$

Here k is the “cycle length”.

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Theorem (Fixed Points)

$$T_{p,g}(1) \leq \sqrt{2p} + 1/2.$$

Existing Results - Fixed Points and Short Cycles

Theorem (Glebsky and Shparlinski, 2010)

$$T_{p,g}(2) \leq C(g) \frac{p}{\log p},$$

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There are no known non-trivial results for $k \geq 4$.

Almost Fixed Points - Preparation

Definition

Let p be a prime, and g be a generator modulo p . Define the function $e_p : \mathbb{Z} \rightarrow \mathbb{C}$ by

$$e_p(x) := \exp\left(\frac{2\pi i x}{p}\right).$$

Define $S_p(a)$ by

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Lemma

Let $a \in \mathbb{Z}$, $a \not\equiv 0 \pmod{p}$. Then

$$S_p(a) \ll p^{1/2} + a.$$

Almost Fixed Points - Main Result

Lemma (Montgomery, 1994)

For any $A \geq 1$ we have

$$\left| M_{p,g}(h) - \frac{2h}{p}(p-1) \right| \ll \Delta_p,$$

where the implied constant does not depend on A , and

$$\Delta_p = p/A + \sum_{a=1}^A (1/A + h/p) |S_p(a)|.$$

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$$M_{p,g}(h) = \frac{2h}{p}(p-1) + O(p^{1/2} + p^{1/4}h^{1/2}).$$

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- The error term Δ_p does not exceed $O(p^{3/4})$.
- For any $\varepsilon > 0$, if $h \geq p^{1/2+\varepsilon}$, then

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which gives an asymptotic formula $M_{p,g}(h) \sim \frac{2h}{p}(p-1)$.

- When $h \leq p^{1/2}$, then Δ_p exceeds the main term, so the result is weak when h is small.

Definition

Define the error Δ_p by

$$\Delta_p(h) := |M_{p,g}(h) - 2h|.$$

Almost Fixed Points - Numerical Analysis ($p \approx 10^9$)

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Figure: Δ_p/p for 20 primes.

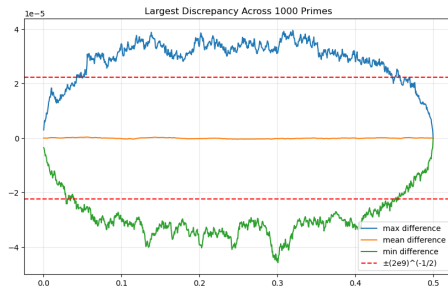


Figure: Max and Min Δ_p/p across 1000 primes.

Almost Fixed Points - Numerical Analysis ($p \approx 10^9$)

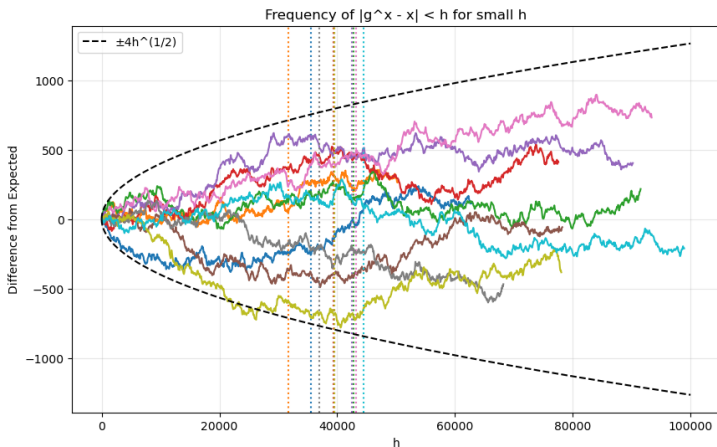


Figure: Δ_p for 10 primes, $h \ll \sqrt{p}$.

Problem 2 - Consecutive Powers in Balls

Definition

Let p be a prime, and $u_0, g \in \{1, \dots, p-1\}$ be elements $\mathbb{Z}_p$ such that g has order t . Let k be a positive integer. Define k -dimensional vectors $\alpha_i \in [0, 1)^k$ by

$$\alpha_i := \frac{1}{p}((u_0 g^i, u_0 g^{i+1}, \dots, u_0 g^{i+k-1}) \pmod{p})$$

for $i = 0, \dots, t-1$.

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for $i = 0, \dots, t-1$.

Definition

For $\mathbf{a} \in [0, 1)^k$ and $r > 0$, define the function

$$N_{p,g}(\mathbf{a}, r) := |\{0 \leq i \leq t-1 : \|\alpha_i - \mathbf{a}\| < r\}|.$$

Consecutive Powers - Discrepancy

To investigate the distribution of vectors in higher dimension, we define the following notion.

Definition

Let N, k be positive integers, and let $S = \{\alpha_1, \alpha_2, \dots, \alpha_N\}$ be a set of N elements in $[0, 1)^k$. For a subset $X \subseteq [0, 1)^k$, let $A(X)$ be the number of elements of S contained in X . Define the **discrepancy** D_N by

$$D_N := \sup_{X \subseteq [0, 1)^k} |A(X)/n - V_k(X)|,$$

where $V_k(X)$ is the volume of X .

- D_N is a measure of 'how far from uniformly distributed' a set is in $[0, 1)^k$.

Definition

A **box** in $[0, 1)^k$ is a set of the form

$$\{\mathbf{x} = (x_1, \dots, x_k) \in [0, 1)^k : a_i \leq x_i \leq b_i \text{ for } i = 1, \dots, k\},$$

for constants $a_1, \dots, a_k, b_1, \dots, b_k$.

Existing Results - Discrepancy in Boxes

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for constants $a_1, \dots, a_k, b_1, \dots, b_k$.

Theorem (Koksma and Szűsz)

Let $\{\mathbf{x}_0, \dots, \mathbf{x}_{n-1}\}$ be points in $[0, 1)^k$. For any integer r ,

$$D_n \ll \frac{1}{r} + \sum_{0 < \|\mathbf{b}\|_\infty \leq r} \frac{1}{h(\mathbf{b})} \left| \frac{1}{n} \sum_{j=0}^{n-1} \exp(2\pi i \mathbf{b} \cdot \mathbf{x}_j) \right|,$$

where the implied constant depends only on k , and

$$h(\mathbf{b}) = \prod_{i=1}^s \max(|b_i|, 1).$$

Theorem

Suppose S is a bound for $\left| \sum_{j=0}^{n-1} \exp(2\pi i \mathbf{b} \cdot \mathbf{x}_j) \right|$ for all $0 < \|\mathbf{b}\|_\infty \leq r$.
Then

$$D_n \ll \frac{1}{r} + \frac{(\log r)^k}{n} S.$$

Existing Results - Discrepancy in Balls

Very little investigation into ball discrepancy had been done previously.

Theorem (Schmidt, 1975)

Let J_n be the discrepancy for k -dimensional balls, and D_n be the discrepancy for boxes. For any $\varepsilon > 0$, we have

$$D_n^{k+\varepsilon} \ll J_n \ll D_n^{1/k},$$

where the implied constant for the lower bound depends on ε .

Consecutive Powers - Main Result

Lemma (Harman, 1998)

For any $s > 0$,

$$|N_{p,g}(\mathbf{a}, r) - tV_k(r)| \ll \Delta,$$

where the implied constant depends only on k , and

$$\begin{aligned} \Delta = & t(r^{k-1}s^{-1} + s^{-k}) \\ & + \sum_{0 < |\mathbf{m}| < s} \left(\frac{1}{s^k} + \min \left(r^k, \frac{r^{(k-1)/2}}{|\mathbf{m}|^{(k+1)/2}} \right) \right) \left| \sum_{n=1}^t e_p(g^n \mathbf{m} \cdot \mathbf{u}) \right|, \end{aligned}$$

where $\mathbf{u} = (u_0, gu_0, \dots, g^{k-1}u_0)$.

Note here the bound is of a similar style to the result used in box discrepancy, however the expression is much more complex.

Consecutive Powers - Main Result

Definition

Let $\mathbf{u} = (u_0, u_0g, \dots, u_0g^{k-1}) \pmod{p}$. Let w be the largest integer such that

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for all $0 < \|\mathbf{m}\| \leq w$.

Theorem

Suppose $g \in \{1, \dots, p-1\}$ has order $t \gg p^{1/2}$. If

$$r \gg \max \left\{ w^{-1}, p^{\frac{1}{2k}} t^{\frac{-1}{k}} \right\},$$

then we have

$$N_{p,g}(\mathbf{a}, r) = tV_k(r) + O \left(tr^{k-1}w^{-1} + t^{\frac{k+1}{k+3}} p^{\frac{1}{k+3}} r^{\frac{k(k+1)}{k+3}} \right).$$

Corollary

Let $\varepsilon > 0$. If

$$r \gg p^\varepsilon \cdot \max \left\{ w^{-1}, p^{\frac{1}{2k}} t^{\frac{-1}{k}} \right\},$$

then

$$N_{p,g}(\mathbf{a}, r) \sim tV_k(r).$$

For almost all g , the condition can be relaxed to just

$$r \gg p^{\frac{1}{2k} + \varepsilon} t^{\frac{-1}{k}}.$$

Numerical Analysis - Ball Centred at Origin

We first look at the discrepancies of balls of radius $1/2$ centred at the origin. We chose g with order $\frac{p-1}{3}$.

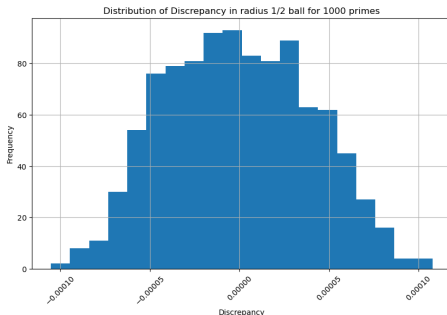


Figure: Distribution of discrepancy for ball of radius $1/2$ centred at origin.

Numerical Analysis - Ball Centred at Origin

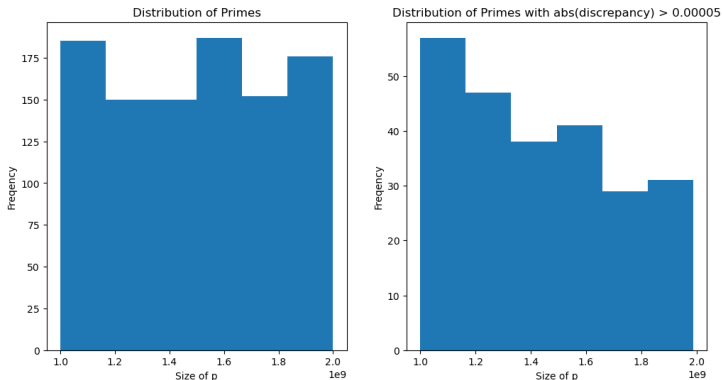


Figure: Distribution of p with absolute discrepancy larger than 0.00005.

Numerical Analysis - Random Balls

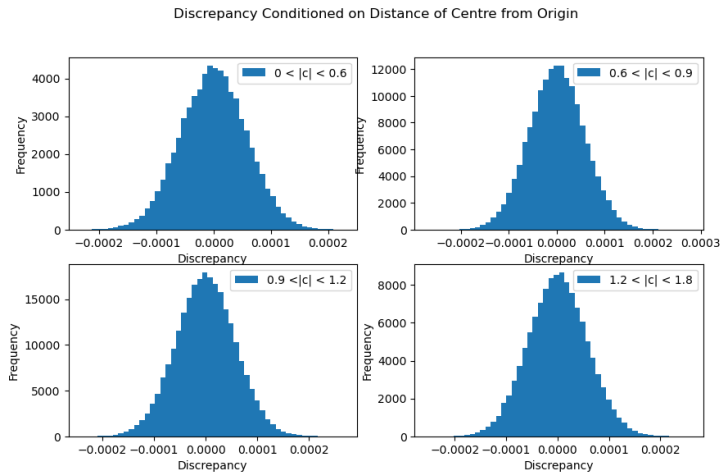


Figure: Distribution of discrepancy for balls of different distances from the origin $r = 0.2$ across 300 primes.

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- Extended the work on the discrepancy of consecutive powers modulo p in k -dimensional boxes to balls, and proved an asymptotic formula for $N_{p,g}(\mathbf{a}, r)$ for large values of r .

Summary and Future Work

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- Numerical results suggest both problems have tighter bounds than the proven results.

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Future Work

- Numerical results suggest both problems have tighter bounds than the proven results.
- Further investigation into results for $M_{p,g}$ for non-primitive elements of $\mathbb{Z}_p$, or for composite modulus.

Thank You! Questions?

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